

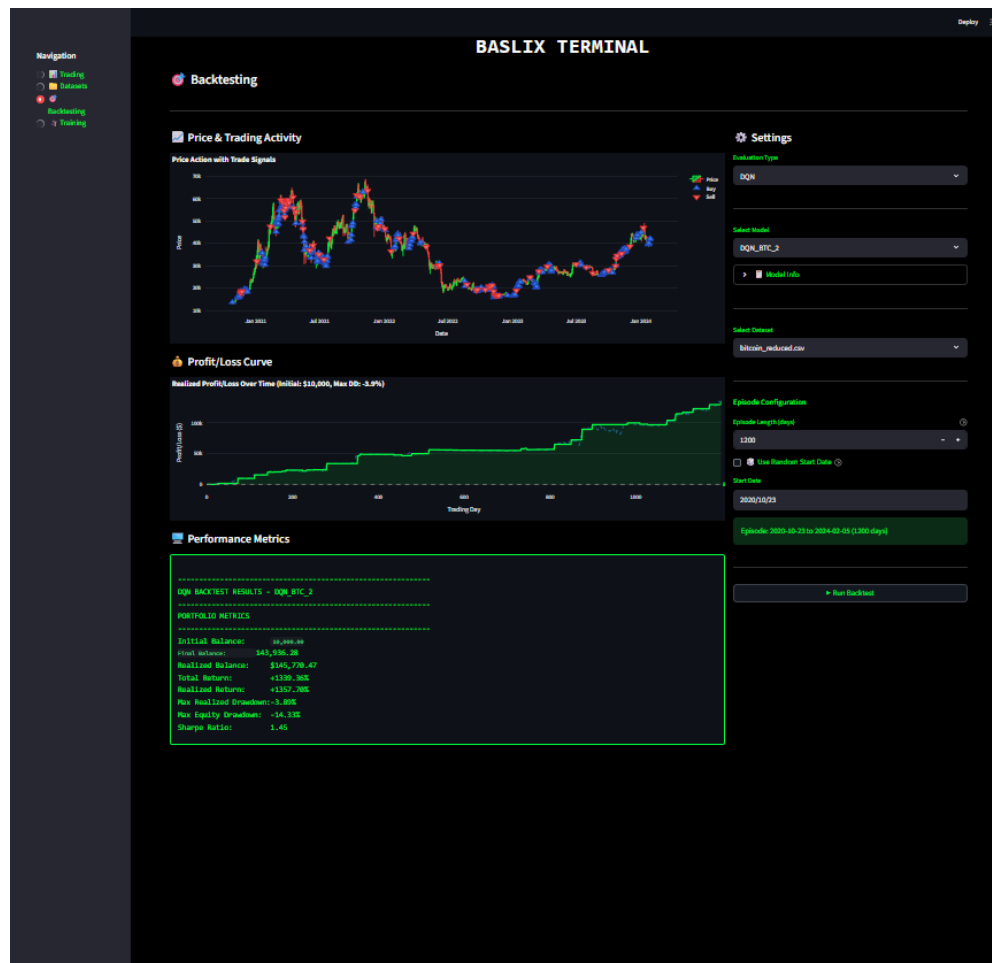
BASLIX Terminal v1: Validation Post-Mortem

Graded A-Level Computer Science NEA (71/75)

Author: *Daniel Dennis (18),*

University of Warwick Physics BSc Student

Project Timeline: *April 2025 - January 2026*



Please note this is a UI screenshot from early tests and is not reflective of real results

This document records a validation flaw found after the submission of this project, and its effect on the results

Overview:

This project was built over a ten-month time period and consists of a fully integrated ML trading research system combining *deep reinforcement learning*, *gradient boosted trees*, and *NLP sentiment analysis*, applied to Gold (2007-2025) and Bitcoin (2015-2025). At the start of development, I lacked financial domain and machine learning knowledge. All this was learnt within the timeline of this project.

This document summarises the architecture, results, and validation issues that I have found, alongside my current plans for a rebuild.

KNOWN VALIDATION ISSUES – READ BEFORE INTERPRETING RESULTS

A **horizon overlap** issue in the LightGBM size capped expanding windows was identified after submission of these results. For a 7-day target, training data was not embargoed for the past 7 days, so the models are **partially fit to outcomes which are unseen**. A 7-day target's label is the return from t to $t+7$, so training rows in the last 7 days of a window have labels computed from prices inside the test period. This is **not traditional feature leakage** but still inflates results.

The problem propagates through the pipeline, so **LightGBM-dependent results are overestimated**. Further testing is required after v2 to test the robustness of the DQN, and if it retains its performance. **The degree of inflation is unknown without a clean rebuild**.

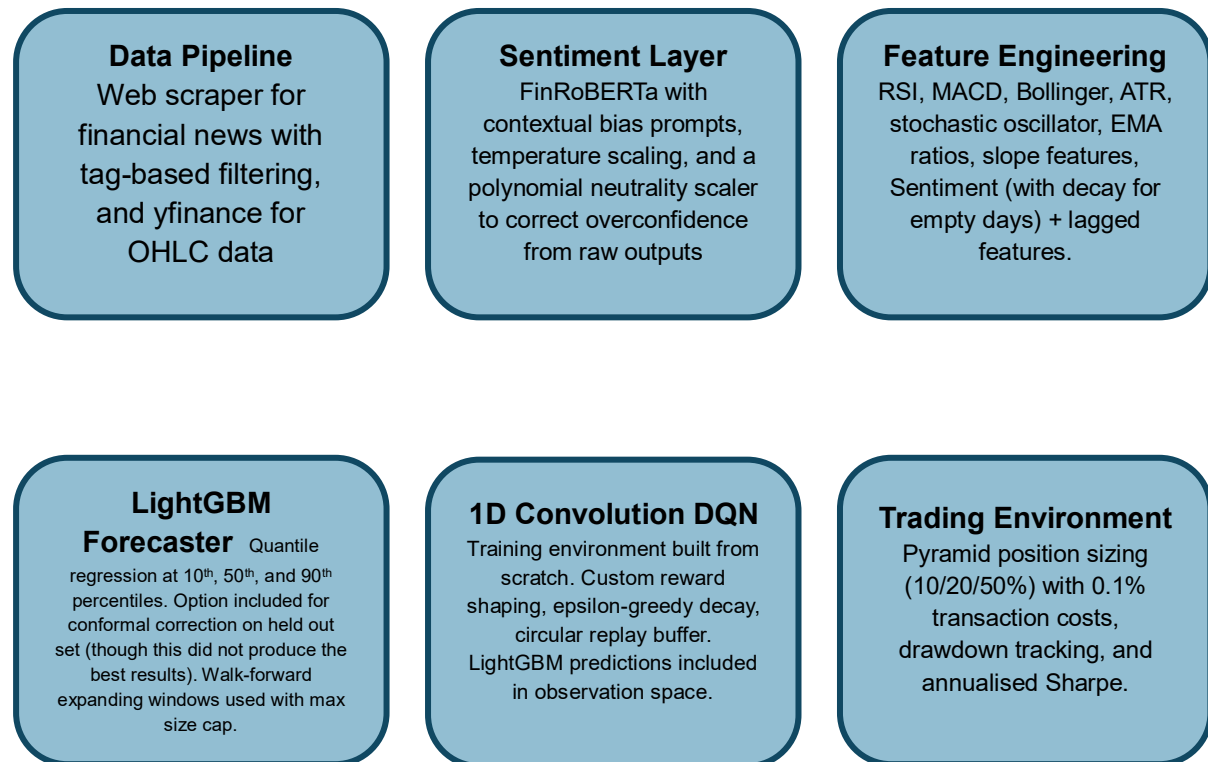
I am flagging this here as I am aware these results are extremely unlikely to hold under real market conditions, and there may still be issues which I have not found yet.

Key Results (as originally reported, now invalid)

Component	Metric	Result
LightGBM (GLD, 7d horizon)	Directional Accuracy	63.4% (Z=18.25)
DQN (BTC-USD, OOS 2021-25)	Total return	+1054%
DQN (BTC-USD, OOS 2021-25)	Sharpe ratio	1.60
DQN (BTC, OOS 2021-25)	Trade win rate	83.2% (n=280)
DQN (GLD, 20 OOS Episodes)	Mean annual return	+45.2% (Z=5.35)

Technical Architecture

The architecture is independent of the validation issue. Contamination is in the capped expanding window scheme, not component design.



Approach and Limitations

Testing methodology:

Three hypothesis tests were conducted at $\alpha=0.05$ using one-tailed Z-tests on out-of-sample data:

1. LightGBM directional accuracy vs 50% random ($Z = 18.25$, $n=4623$)
2. DQN trade win rate vs 50% random ($Z=11.1$, $n=280$)
3. DQN annual returns vs zero ($Z=5.35$, $n=20$ episodes)

All tests reject the null hypothesis by a comfortable margin, however with the discovery of the leakage, this is invalid evidence.

Leakage Discovery:

While testing the model, more frequent retraining seemed to have a positive effect on the effectiveness of the DQN, but to a more extreme extent that seemed appropriate. This put the retrain window logic under greater scrutiny, leading to the discovery of the leakage issue.

Limitations:

- Horizon overlap leakage in LightGBM expanding windows propagates to DQN results
- The sample size for Test 3 is limited despite the large number of possible episodes within the 4-year OOS period. Since many randomly sampled 252-day episodes overlap substantially, the effective number of independent observations is reduced, weakening the standard hypothesis test.
- Overlapping predictions and trades violate the independent observation assumption of the Z-test
- No regime-stratified analysis (many different regimes are included within the single OOS period)
- For test 1, with a 7-day target sampled daily, consecutive labels will overlap, reducing the effective number of independent observations to $4623/7$ which is ~ 660 . Recomputing at this effective n gives $Z \approx 18.25/\sqrt{7} \approx 6.9$, which is still significant but far from the reported figure.
- The 50% baseline used for directional accuracy also assumes an even split of up/down days. In future tests, the actual up/down split should be used as a baseline, and gold mostly trended up over much of 2007-2025.
- The DQN Agent is long-only, so its performance might partly reflect the general uptrend in BTC-USD over the training/testing periods.

The Rebuild

- Proper embargo windows for LightGBM
- Replace yfinance with IB TWS API
- Make use of DAG feature engineering with topological sort and cycle detection
- VAE for market regime detection
- Add a neural network to optimise the parameters of a Monte-Carlo simulation

NEA Document Design, Analysis, and Evaluation sections as well as specific code modules are available on request.